

50,000 MIMIC-IV discharge summaries
312 shards, stratified sample
the text never leaves the compute volume

46 ICD-9 codes per note

written by human coders for billing,
before any model saw the text

the codes are the answer key in both tests

Gemma-2-2B, layer 16 residual stream
2,304-d, mean-centred, 15.2 M tokens

Why these labels: they were written by clinicians
for billing, before this study existed, so nothing in
the pipeline could have shaped them.

1 Feature sources – anything that maps a note to per-token values

trained on this corpus

ReLU + L1
SAE
k = 18,432

JumpReLU
SAE
k = 18,432

off-the-shelf

GemmaScope
SAE
k = 16,384

no labels, no learning

PCA
directions
k = 2,304

random,
cov-matched
k = 18,432

surface text

TF-IDF
n-grams
k = 10,000

built from the ICD-9 labels – the only three sources that see the codes

diff-in-means
(LDA)
k = 46

LR probe
directions
k = 46

keyword
indicators
k = 46

2 One interface for every source

max-pool each feature over the note's tokens,
giving one [notes × k] matrix in the same format for all nine

3 The audit – one code path, every source

select one feature per code, chosen on shards 0-30
score point-biserial r on shards 281-311, BH-FDR at $q = 0.05$
report on-target |r|, off-target leakage, specificity ratio
*the two splits never overlap, so the best of 18,432 features is never
scored on the notes that chose it*

one feature per code, per source

4a Concordance test

can a judge who never sees the code recover
it from the feature's explanation alone?

the feature's top-activating contexts
pulled from held-out notes, shards 281-311
no note text ever leaves the volume

explainer LLM writes one sentence
sees the contexts only – no code, no |r|
Claude Sonnet 4.6

judge LLM picks 1 of 9 options
8 codes from the same ICD-9 chapter, plus 'none'

hit@1, against 11.1 % chance
two judge families, three disjoint |r| bands

4b Causal ablation test

does the model actually use the feature,
or does it only correlate with the code?

remove the feature at layer 16
SAE latent: set it to 0, or to its dataset mean
plain direction d : $x' = x - (x \cdot d) d$

re-run the remaining layers
cross-entropy on the note's last 25 % of tokens
three passes: clean, reconstructed, ablated

split the 4,911 held-out notes in two
notes carrying the code, vs notes without it

does loss rise more on code-positive notes?
Cliff's δ , Mann-Whitney U, BH across targets

Every one of the nine sources is run through both tests, so each SAE number is read against a
random-direction floor and a keyword ceiling measured on the identical pipeline – not against nothing.